

On the Implicit Function Theorem and Consequences: Part I

Dr Francois Ebobisse

October 2024

1 Introduction

Let $F : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a function. Consider the equation

$$F(x, y) = 0. \quad (1)$$

In general, to a value of x there can correspond one or more values of y for which the equation (1) holds. In this section, we want to study the conditions under which (1) defines y as a function of x , at least for some values of these variables.

Let us first consider some examples which illustrate various situations that can occur.

Example 1.1

- (i) Let $F(x, y) = y^5 - x$. In this case the unique function defined by (1) is $f(x) = \sqrt[5]{x}$.
- (ii) Let $F(x, y) = x^2 + y^2 - 4$. The equation (1) defines in this case two different functions $y = f_1(x)$ and $y = f_2(x)$ defined on the interval $I = [-2, 2]$ by

$$f_1(x) = -\sqrt{4 - x^2} \quad \text{and} \quad f_2(x) = \sqrt{4 - x^2} \quad \forall x \in [-2, 2].$$

- (iii) $F(x, y) = x^2 + y^2 + 4$. For any $(x, y) \in \mathbb{R}^2$, we have $F(x, y) \neq 0$. Therefore for any interval I of $\mathbb{R}$, there is no function $f : I \rightarrow \mathbb{R}$ such that $F(x, f(x)) = 0$ for $x \in I$.
- (iv) Let $F(x, y) = x^3 + y^3 + x^2y - 3y^2$. It is easy to see that for any $x > \sqrt{3}$, the function $y \rightarrow F(x, y)$ is strictly increasing. We also have that

$$\lim_{y \rightarrow -\infty} f(x) = -\infty \quad \text{and} \quad \lim_{y \rightarrow \infty} F(x, y) = +\infty.$$

Therefore, the equation (1) defines for any $x > \sqrt{3}$ a unique y . That is, (1) defines a unique function $f : (\sqrt{3}, +\infty) \rightarrow \mathbb{R}$ such that $F(x, f(x)) = 0$ for any $x > \sqrt{3}$.

Notice that the equation $F(0, y) = 0$ has two solutions $y = 0$ and $y = 3$ and that the equation $F(1, y) = 0$ has three solutions $y = 1, 1 \pm \sqrt{2}$.

The following theorem gives the condition under which (1) defines y as a function of x .

2 Dini's theorem for two variables functions

Theorem 2.1 *Let $\Omega \subset \mathbb{R}^2$ be an open set and let $F : \Omega \rightarrow \mathbb{R}$ be a continuous function such the partial derivative F_y is continuous in Ω . If (a, b) is a point in Ω such that*

$$F(a, b) = 0 \quad \text{and} \quad F_y(a, b) \neq 0$$

Then there exist two numbers $\delta > 0$ and $\beta > 0$ such that $(a - \delta, a + \delta) \times (b - \beta, b + \beta) \subset \Omega$ and the equation $F(x, y) = 0$ for any $(x, y) \in (a - \delta, a + \delta) \times (b - \beta, b + \beta)$ defines implicitly a unique function

$$f : (a - \delta, a + \delta) \rightarrow (b - \beta, b + \beta),$$

that is, a function such that

$$F(x, f(x)) = 0 \quad \forall x \in (a - \delta, a + \delta).$$

Moreover, f is continuous at a and $f(a) = b$. If F is of class $\mathcal{C}^1$ in Ω , then the function f is of class $\mathcal{C}^1$ in $(a - \delta, a + \delta)$ and

$$f'(x) = -\frac{F_x(x, f(x))}{F_y(x, f(x))} \quad \forall x \in (a - \delta, a + \delta).$$

In general, if F is of class $\mathcal{C}^k$ in Ω for some $k \in \mathbb{N}$, then the function f is of class $\mathcal{C}^k$ in $(a - \delta, a + \delta)$.

Proof: Let us assume that $F_y(a, b) > 0$. Hence, from the continuity of F_y , there exists a number $\delta_1 > 0$, $\beta > 0$ such that

$$F_y(x, y) > 0 \quad \text{for all } (x, y) \in \Omega \text{ such that } |x - a| \leq \delta_1 \text{ and } |y - b| \leq \beta.$$

Hence, the single variable function $y \rightarrow F(a, y)$ is strictly increasing in $[b - \beta, b + \beta]$. Since, $F(a, b) = 0$, it follows that

$$F(a, b - \beta) < 0 \quad \text{and} \quad F(a, b + \beta) > 0.$$

Now since the function $x \rightarrow F(x, b - \beta)$ and $x \rightarrow F(x, b + \beta)$ are continuous, there exists $\delta > 0$ (and less than δ_1) such that

$$F(x, b - \beta) < 0 \quad \text{and} \quad F(x, b + \beta) > 0 \quad \forall x \in (a - \delta, a + \delta).$$

Hence, for any $x \in (a - \delta, a + \delta)$, the function $y \in [b - \beta, b + \beta] \rightarrow F(x, y)$ is strictly increasing takes negative and positive values at the end points of the interval $[b - \beta, b + \beta]$. Thus, by the intermediate value theorem, for any $x \in (a - \delta, a + \delta)$, there exists a unique $y \in (b - \beta, b + \beta)$ such that $F(x, y) = 0$.

Therefore we have proved the existence of a unique implicit function $f : (a - \delta, a + \delta) \rightarrow (b - \beta, b + \beta)$ and by construction $b = f(a)$.

Now let us prove that f is continuous in $(a - \delta, a + \delta)$. To simplify the notations set $I = (a - \delta, a + \delta)$ and $J := (b - \beta, b + \beta)$.

Fix $x_0 \in I$ and let us prove that f is continuous at x_0 . Let $\varepsilon > 0$. We want to prove that

$$|f(x) - f(x_0)| < \varepsilon \quad \forall x \in I \text{ with } x \text{ sufficiently close to } x_0. \quad (2)$$

Assume that ε is sufficiently small such that $[f(x_0) - \varepsilon, f(x_0) + \varepsilon] \subset J$ (this is possible because $f(x_0) \in J$ and J is open).

Since $F(x_0, f(x_0)) = 0$ and the function $y \rightarrow F(x_0, y)$ is strictly increasing in $[f(x_0) - \varepsilon, f(x_0) + \varepsilon]$, we have that

$$F(x_0, f(x_0) - \varepsilon) < 0 < F(x_0, f(x_0) + \varepsilon).$$

Now arguing as above, we have some number $\delta_2 > 0$ ($\delta_2 < \delta$) such that

$$F(x, f(x_0) - \varepsilon) < 0 < F(x, f(x_0) + \varepsilon) \quad \forall x \in I \text{ with } |x - x_0| < \delta_2.$$

Now since, $F(x, f(x)) = 0$ for any $x \in I$ and the function $y \in F(x, y)$ is strictly increasing, it follows that

$$f(x_0) - \varepsilon < f(x) < f(x_0) + \varepsilon \quad \forall x \in I \text{ with } |x - x_0| < \delta_2,$$

which proves (2) for ε small and hence, (2) also holds for ε large.

Now assume that F is of class $\mathcal{C}^1$ in Ω . Following the proof of the first part, we have that

$$F_y(x, y) \neq 0 \quad \forall (x, y) \in R := (a - \delta, a + \delta) \times (b - \beta, b + \beta),$$

Let us fix x and $h \neq 0$ such that $x, x + h \in (a - \delta, a + \delta)$. From the definition of the function f and the Mean value theorem, it follows that

$$\begin{aligned} 0 &= F(x + h, f(x + h)) - F(x, f(x)) \\ &= [F(x + h, f(x + h)) - F(x, f(x + h))] + [F(x, f(x + h)) - F(x, f(x))] \\ &= F_x(s, f(x + h))h + F_y(x, t)[f(x + h) - f(x)] \end{aligned}$$

where s and t are two numbers between x and $x + h$ and $f(x)$ and $f(x + h)$ respectively. Therefore

$$\frac{f(x + h) - f(x)}{h} = -\frac{F_x(s, f(x + h))}{F_y(x, t)}.$$

Now when h goes to 0, we get

$$(s, f(x + h)) \rightarrow (x, f(x)), \quad (x, t) \rightarrow (x, f(x)), \quad F_x(s, f(x + h)) \rightarrow F_x(x, f(x)) \text{ and } F_y(x, t) \rightarrow F_y(x, f(x)).$$

Thus

$$\lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h} = -\lim_{h \rightarrow 0} \frac{F_x(s, f(x + h))}{F_y(x, t)} = -\frac{F_x(x, f(x))}{F_y(x, f(x))}.$$

Therefore

$$f'(x) = -\frac{F_x(x, f(x))}{F_y(x, f(x))}.$$

Now assume that F is of class $\mathcal{C}^k$ in Ω with $k \geq 2$ integer and we want to prove that f is of class $\mathcal{C}^k$ in $(a - \delta, a + \delta)$. By induction, we assume that if F is of class $\mathcal{C}^{k-1}$ in Ω then f is of class $\mathcal{C}^{k-1}$ in $(a - \delta, a + \delta)$.

Now having F of class f is of class $\mathcal{C}^k$ in Ω , F is also of class $\mathcal{C}^{k-1}$. Therefore, f is of class $\mathcal{C}^{k-1}$ in $(a - \delta, a + \delta)$. Now from

$$f'(x) = -\frac{F_x(x, f(x))}{F_y(x, f(x))} \quad \forall x \in (a - \delta, a + \delta),$$

it follows that f' is of class $\mathcal{C}^{k-1}$ in $(a - \delta, a + \delta)$. That is, f is of class $\mathcal{C}^k$ in $(a - \delta, a + \delta)$. ■

Remark 2.2 If the assumptions F_y continuous and $F_y(a, b) \neq 0$ are replaced by F_x and $F_x(a, b)$ respectively, then the equation $F(x, y) = 0$ defines implicitly a function $x = g(y)$ defines in a neighborhood of b .

Exercise 2.3 Let $\Omega \subset \mathbb{R}^2$ be an open set. Let $F : \Omega \rightarrow \mathbb{R}$ be a function of class $\mathcal{C}^1$ and let $Z := \{(x, y) \in \Omega : f(x, y) = 0\}$. Prove that if $(a, b) \in Z$ is such that $\nabla f(a, b) \neq (0, 0)$, then in a neighborhood of (a, b) the set Z coincides with a simple regular curve whose tangent line at (a, b) has equation: $F_x(a, b)(x - a) + F_y(a, b)(y - b) = 0$.

Exercise 2.4 Let $F(x, y) := xy + \ln(xy) - 1$. Show that around the point $(1, 1)$, the equation $F(x, y) = 0$ defines implicitly a function $y = f(x)$.

3 Dini's theorem for functions of k variables ($k \geq 3$)

In this section we present Dini's theorem for functions of at least three variables

Theorem 3.1 Let Ω be an open subset of $\mathbb{R}^{n+1}$ with $n \geq 2$. Let $x = (x_1, \dots, x_n) \in \mathbb{R}^n$ and $y \in \mathbb{R}$ and let $F(x, y) = F(x_1, \dots, x_n, y)$ be a real-valued function continuous together with its partial derivative F_y in Ω . If $(a, b) = (a_1, \dots, a_n, b) \in \Omega$ is such that

$$F(a, b) = 0 \quad \text{and} \quad F_y(a, b) \neq 0,$$

then, there exist $\delta, \beta > 0$ and a unique function

$$f : B(a, \delta) \rightarrow (b - \beta, b + \beta)$$

such that

$$\forall (x, y) \in B(a, \delta) \times (b - \beta, b + \beta) \quad F(x, y) = 0 \quad \Leftrightarrow \quad y = f(x). \quad (3)$$

In particular, $f(a) = b$ since $F(a, b) = 0$. Moreover, f is continuous.

If F is of class $\mathcal{C}^1$ in Ω , then the function f is of class $\mathcal{C}^1$ in $B(a, \delta)$ and

$$\frac{\partial f}{\partial x_i}(x) = -\frac{F_{x_i}(x, f(x))}{F_y(x, f(x))} \quad \forall x \in B(a, \delta).$$

In general, if f is of class $\mathcal{C}^k$ in Ω for some $k \in \mathbb{N}$, then the function f is of class $\mathcal{C}^k$ in $B(a, \delta)$.

Proof: The proof of the existence of the implicit function can be obtained following the argument used in the proof of Theorem 2.1, based on the monotonicity of the function $y \rightarrow F(x, y)$ and on the continuity of the function $x \rightarrow F(x, y)$. In this proof, we use a different argument based on the method of successive approximations.

For any $(x, y) \in \Omega$ we set

$$\Gamma(x, y) := y - b - \frac{F(x, y)}{F_y(a, b)}.$$

Let us first notice that (3) is equivalent to

$$f(x) = b + \Gamma(x, f(x)) \quad \forall x \in B(a, \delta). \quad (4)$$

In other words the implicit function f is obtained as fixed point of some mapping. So, the strategy is to obtain the function f through the contraction mapping theorem, which requires a contraction mapping on a complete metric space. So, let us prove that (4) holds.

Since Ω is open, Γ_y is continuous and $\Gamma_y(a, b) = 0$, there exist $\alpha > 0$ and $\beta > 0$ such that

$$B[a, \alpha] \times [b - \beta, b + \beta] \subset \Omega \quad \text{and} \quad |\Gamma_y(x, y)| \leq \frac{1}{2} \quad \forall (x, y) \in B[a, \alpha] \times [b - \beta, b + \beta]. \quad (5)$$

Since $\Gamma(a, b) = 0$, it is not restrictive to assume that α is also such that

$$|\Gamma(x, b)| \leq \frac{\beta}{2} \quad \forall x \in B[a, \alpha]. \quad (6)$$

We will assume that $F_y(a, b) > 0$ and hence, by the continuity of F_y , we can also further assume that, α and β are such that

$$F_y(x, y) > 0 \quad \forall (x, y) \in B[a, \alpha] \times [b - \beta, b + \beta]. \quad (7)$$

So, to summarize, we found $\alpha > 0$ and $\beta > 0$ such that

$$\left. \begin{array}{l} B[a, \alpha] \times [b - \beta, b + \beta] \subset \Omega, \\ |\Gamma_y(x, y)| \leq \frac{1}{2}, \quad |\Gamma(x, b)| \leq \frac{\beta}{2}, \\ F_y(x, y) > 0. \end{array} \right\} \quad \forall (x, y) \in B[a, \alpha] \times [b - \beta, b + \beta]$$

We set

$$X := \mathcal{C}(B[a, \alpha], [b - \beta, b + \beta]) \text{ equipped with } d_\infty(f, g) := \sup_{x \in B[a, \alpha]} |f(x) - g(x)| \quad \forall f, g \in X.$$

Claim 1: (X, d_∞) is a complete metric space. In fact, since $(\mathcal{C}(B[a, \alpha], \mathbb{R}), d_\infty)$ is a complete metric spaces (from 3MS and our first lectures on the revision on metric spaces), it is sufficient show that X is a closed subset of $(\mathcal{C}(B[a, \alpha], \mathbb{R}), d_\infty)$. In fact, let (f_n) be a sequence in X which converges uniformly to f . Let us show that $f \in X$. Clearly $f \in \mathcal{C}(B[a, \alpha], \mathbb{R})$. Now, since (f_n) converges pointwise to f , it follows that

$$b - \beta \leq f_n(x) \leq b + \beta \quad \forall n \in \mathbb{N}, \quad \forall x \in B[a, \alpha] \quad \underbrace{\Rightarrow}_{n \rightarrow \infty} \quad b - \beta \leq f(x) \leq b + \beta \quad \forall x \in B[a, \alpha].$$

Hence, $f \in X$.

Now, consider the mapping $\Psi : X \rightarrow X$ defined by

$$\Psi(f)(x) := b + \Gamma(x, f(x)) \quad \forall x \in B[a, \alpha], \quad \forall f \in X.$$

We want to show that the mapping Ψ is a contraction.

Claim 2: The mapping Ψ is well-defined. Let $f \in X$. Clearly, the function $\Psi(f)$ is continuous as a composition of continuous function. Now, let us show that $\Psi(f)(x) \in [b - \beta, b + \beta]$ for every $x \in B[a, \alpha]$. In fact,

$$\begin{aligned} |\Psi(f)(x) - b| &= |\Gamma(x, f(x))| \leq |\Gamma(x, b)| + |\Gamma(x, f(x)) - \Gamma(x, b)| \\ &= |\Gamma(x, b)| + |\Gamma_y(x, t)| |f(x) - b| \quad (\text{Mean Value Theorem, } t \text{ between } f(x) \text{ and } b) \\ &\leq \frac{\beta}{2} + \frac{\beta}{2} = \beta \quad (\text{because } |\Gamma(x, b)| \leq \frac{\beta}{2}, \quad |\Gamma_y(x, t)| \leq \frac{1}{2} \text{ and } |f(x) - b| \leq \beta) \end{aligned}$$

Hence, $\Psi(f)(x) \in [b - \beta, b + \beta]$.

Claim 3: The mapping Ψ is a contraction. Let $f, g \in X$ and let $x \in B[a, \delta]$. Then,

$$\begin{aligned} |\Psi(f)(x) - \Psi(g)(x)| &= |(b + \Gamma(x, f(x))) - (b + \Gamma(x, g(x)))| = |\Gamma(x, f(x)) - \Gamma(x, g(x))| \\ &= |\Gamma_y(x, s)| |f(x) - g(x)| \quad (\text{Mean Value Theorem, } s \text{ between } f(x) \text{ and } g(x)) \\ &\leq \frac{1}{2} |f(x) - g(x)| \quad (\text{because } |\Gamma_y(x, s)| \leq \frac{1}{2}) \\ &\leq \frac{1}{2} d_\infty(f, g). \end{aligned}$$

Therefore,

$$d_\infty(\Psi(f), \Psi(g)) = \sup_{x \in B[a, \alpha]} |\Psi(f)(x) - \Psi(g)(x)| \leq \frac{1}{2} d_\infty(f, g).$$

Hence, the mapping Ψ is indeed a contraction.

Thus, by the contraction mapping theorem, there exists a unique $f \in X$ such that

$$\Psi(f) = f \quad \Leftrightarrow \quad (4).$$

So, we found $\alpha, \beta > 0$ and a unique continuous function $f : B[a, \alpha] \rightarrow [b - \beta, b + \beta]$ such that $F(x, f(x)) = 0$ for every $x \in B[a, \alpha]$. Precisely, from the monotonicity property in (7), it follows that

$$\forall (x, y) \in B[a, \alpha] \times [b - \beta, b + \beta] \quad F(x, y) = 0 \quad \Leftrightarrow \quad y = f(x).$$

In particular, $f(a) = b$ since $F(a, b) = 0$ by assumption.

Notice that the statement of the theorem requires an open ball $B(a, \delta)$ and an open interval $(b - \beta, b + \beta)$.

To this aim, we use the continuity of the function f as follows. Given that $f(a) = b$, we exploit the continuity of f at a , to get $\delta > 0$ such that

$$f(B(a, \delta)) \subset (b - \beta, b + \beta).$$

It is not restrictive to assume that $\delta < \alpha$ (otherwise, we replace δ with the number $\min(\alpha, \delta)$). Hence,

$$f : B(a, \delta) \rightarrow (b - \beta, b + \beta)$$

and this completes the proof of the first part of the theorem.

Claim 4: Now, let assume that F is of class C^1 , that is, all the partial derivatives F_{x_i} and F_y are continuous and let us show that the implicit functions f also has its partial derivatives f_{x_i} continuous.

Now, let x and $h \neq 0$ be such that $x, x+h \in B(a, \delta_1)$. Let us fix $i \in \{1, \dots, n\}$. Let e_i be the i th vector in the canonic basis of $\mathbb{R}^n$. We have from the Mean Value Theorem that

$$\begin{aligned} 0 &= F(x + he_i, f(x + he_i)) - F(x, f(x)) = [F(x + he_i, f(x + he_i)) - F(x, f(x + he_i))] \\ &\quad + [F(x, f(x + he_i)) - F(x, f(x))] = F_{x_i}(s, f(x + he_i))h + F_y(x, t)[f(x + he_i) - f(x)] \end{aligned}$$

for some point s between $x + he_i$ and x and some number t between $f(x + he_i)$ and $f(x)$. Therefore

$$\frac{f(x + he_i) - f(x)}{h} = -\frac{F_{x_i}(s, f(x + he_i))}{F_y(x, t)}. \quad (8)$$

So, passing to the limit in (8) when h goes to 0 we get

$$\frac{\partial f}{\partial x_i}(x) = -\frac{F_{x_i}(x, f(x))}{F_y(x, f(x))} \quad \forall x \in B(a, \delta_1) \quad (9)$$

and we also have that f_{x_i} is continuous. Therefore, the function f is of class $\mathcal{C}^1$ in $B(a, \delta_1)$. Moreover, if F is of class $\mathcal{C}^k$ ($k \geq 2$), it follows from (9) that f is also of class $\mathcal{C}^k$ and this completes the proof the theorem. ■

Example 3.2 Consider the function

$$F(x, y, z) := x + \sin y - 2y - z + \cos z - 2.$$

- (a) Show that the equation $F(x, y, z) = 0$ defines a function of class $\mathcal{C}^\infty$ around the point $(1, 0, 0)$.
- (b) Write down the cartesian equation of the tangent plane to the graph of the function f at the point $(1, 0, 0)$.

Solution:

- (a) The function F is of class $\mathcal{C}^\infty$. It is easy to see that

$$F(1, 0, 0) = 0 \quad \text{and} \quad F_z(x, y, z) = -1 - \sin z \quad \Rightarrow \quad F_z(1, 0, 0) = -1 \neq 0.$$

Therefore, by the implicit function theorem (IFT), there exist $\delta > 0$ and $\beta > 0$ and a function

$$f : B((1, 0), \delta) \rightarrow (-\beta, \beta) \text{ of class } \mathcal{C}^\infty$$

such that

$$\forall (x, y) \in B((1, 0), \delta), \forall z \in (-\beta, \beta), \quad F(x, y, z) = 0 \quad \Leftrightarrow \quad z = f(x, y).$$

- (b) The tangent plane to the graph of f around the point $(1, 0, 0)$ has cartesian equation (from (2AC))

$$z = f(1, 0) + f_x(1, 0)(x - 1) + f_y(1, 0)(y - 0).$$

We have from Theorem 3.1 that

$$f_x(x, y) := -\frac{F_x(x, y, f(x, y))}{F_z(x, y, f(x, y))} \quad \text{and} \quad f_y(x, y) := -\frac{F_y(x, y, f(x, y))}{F_z(x, y, f(x, y))}.$$

Notice that

$$F_x(x, y, z) = 1 \text{ and } F_y(x, y, z) = \cos y - 2 \quad \Rightarrow \quad F_x(1, 0, 0) = 1 \text{ and } F_y(1, 0, 0) = -1.$$

Hence,

$$f_x(1, 0) = -\frac{1}{-1} = 1 \quad \text{and} \quad f_y(1, 0) = -\frac{-1}{-1} = -1.$$

So, knowing that $f(1, 0) = 0$, we get the cartesian equation

$$z = 0 + 1 \times (x - 1) - 1 \times (y - 0) \quad \Leftrightarrow \quad \boxed{z = x - y - 1}.$$

The next lecture will be on the implicit function theorem for vector valued functions.